

POWER AND ENERGY IN UGANDA.

Power plays a significant role in the Country's socio-economic development while providing the necessary infrastructural economic base for Uganda's attractiveness for investment in the various sectors particularly industry, commerce and agriculture.

The energy sector in Uganda comprises of both traditional and conventional energy sources that are either locally produced or imported. The most prevalent used energy sources that are locally produced are wood fuel and charcoal.

Other energy sources that are used include petroleum products, electricity and renewable energy sources.

CURRENT STATUS OF THE ENERGY SECTOR.

- Uganda's current energy balance comprises 92% biomass, 7% fossil fuels and 1% electricity.
- Energy from biomass contributes about 92% of the energy used in Uganda.
- Most of the biomass energy is from wood which is consumed in form of firewood and charcoal.
- The critical source of energy for Uganda is electricity which is used for Industrial and commercial production. This electricity is generated from HEP supplemented by thermal power and solar energy.
- Access to HEP has been increasing over time.
- Hydro electric power is a popular source of energy in the urban centres and industries.
- Hydro electric power production, distribution and marketing has been privatized.
- There is increased use of solar electricity in most parts of rural Uganda because of the high tariffs and limited connectios of HEP.
- Payments for Hydro electricity has been digitized using pre paid Yaka gadgets to reduce on power theft.
- There is increased production of hydro electricity to hit the targeted megawatts of about 5300 MW by 2030.
- Currently HEP contributes 745MW produced from Bujjagali, Nalubaale, Isimba all on River Nile and many other mini hydro power dams.
- Rural electrification program is going on.
- There is increased installation of solar energy panels in most rural areas.
- In 2019 Uganda launched the largest solar plant producing 20MW at Kabulasoke in Gomba.
- The sector is developing.
- Petroleum production is in advanced stage in the Albertine Graben.

- There is increased use of biogas in areas like Mbarara, Kabale and Nakasongola.
- Geo thermal energy is being piloted at Ssempaya Hotsprings in Bundibugyo.
- The mini hydro power plants account for 38.84mw.
- There is cogeneration to the national grid by Biomass at Kakira Sugar Factory in Jinja of about 12MW, and 5MW by Kinyara Sugar Factory in Masindi.
- Thermal energy generates 170MW to the national grid generated from Namanve in Mukono, Tororo and Kiira plant in Wakiso.
- Sale for all petroleum products has been on increase as a result of increase in vehicles and motorcycles.
- Other potentials like geothermal energy have not been exploited.

ENERGY SOURCES IN UGANDA.

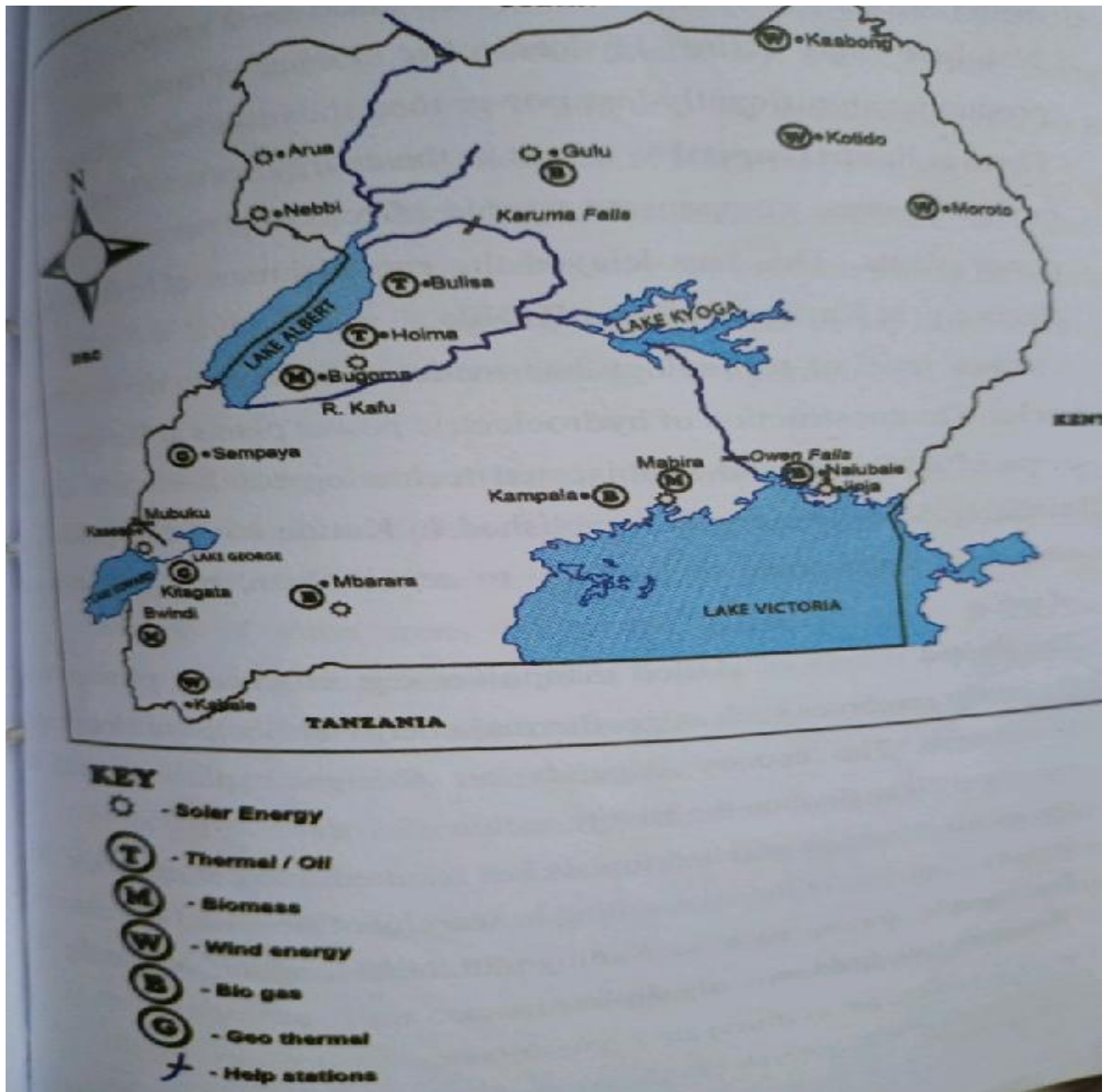
- 1) Biomass for example firewood from Mabira Forest in Buikwe, Budongo Forest in Masindi, charcoal from forests etc. Other forms of biomass include coffee husks, wood husks, sugarcane residues in Kakira Jinja and Lugazi in Buikwe.
- 2) Petroleum energy used in major cities and towns such as Kampala, Jinja, Arua and Gulu etc. All the county's petroleum products are imported. It's used in industry, agriculture, transport and domestic use.
- 3) Hydro electric power generated on various dams like Bujjagali, Nalubaale and Kiira all on River Nile etc
- 4) Thermal energy/ diesel generators at Namanve, Buliisa, Kalangala and Hoima.
- 5) Biogas (from plants and animal wastes) for example in Mbarara , Wakiso, Kampala etc.
- 6) Solar energy for example solar panels like PV solar farm in Tororo, Kabulasoke Solar plant in Gomba and many private panels installed in households, health centres, schools and water pumping stations in Soroti, Kampala, Mbale Mbarara Gulu etc.
- 7) Wind energy in Moroto, Kotido, Kaabong, Nakapiripiriti and Abim.

The following table shows Uganda's most important power stations.

PLANT/SOURCE	DISTRICT	CAPACITY(MW)
Kiira hydro power station	Jinja	200
Nalubaale hydro power station	Jinja	180
Isimba hydro station	Kayunga/Kamuli	183
Bujagaali	Jinja	100
Jacobsen Namanve Thermal power plant	Mukono	50
Electro Max Thermal power plant	Kampala	20

Kakira sugar works cogeneration	Kakira Jinja	20
Kinyara Sugar works cogeneration	Masindi	7
Kilembe mines Ltd small hydro power plant	Kasese	5
Bugoye small hydro power plant	Kasese	13
Ishasha small hydro power plant	Kanungu	6.5
Mpanga small hydro power plant	Kamwenge	18

A SKETCH MAP OF UGANDA SHOWING ENERGY SOURCES.



REASONS FOR THE LOW LEVEL OF DEVELOPMENT OF THE ENERGY SECTOR IN UGANDA

- Fluctuations in water levels due to prolonged drought coupled with increased discharges from Lake Victoria. As a result this has affected power supply at Nalubaale and Kiira Hydro electric power plants at Jinja.

- Limited capital to invest in the energy sector. There is less capital required to buy generators, compensate people affected by the establishment of power plants. This has delayed the completion of hydro power plants like Karuma Dam in Kiryandongo.
- Low levels of technology has made it difficult to develop the energy sector. The construction of Karuma dam in Kiryandongo , Isimba Power Dam in Kayunga and Mubuku hydro power plant delayed because they required advanced and sophisticated technology which was not locally available. Many windmills established in Kotido have collapsed due to inappropriate technology to repair or replace them. Many have gone out of use.
- There is limited supply of skilled labour like engineers, planners to tap the available energy resources such as geothermal energy at Sempaya Hot Springs in Bundibugyo. The country depends on foreign expatriates with the relevant skills to develop the energy sector for example the construction of Karuma Power station in Kiryandongo is being done by Energy Infratech an Indian company.
- Encroachment on gazetted forests has resulted into deforestation. This has led to depletion of forests hence causing a scarcity of firewood and high cost of charcoal in urban centres like Kampala where a sack is almost costing between 5000 -70000 Uganda shillings. The forests depleted include Katuugo Forest in Nakasongola, Nawandagi Forest in Butambala etc.
- The high levels of corruption has limited the development of the energy sector. The corruption involving AES Nile power, which was responsible for the building of Bujagali Dam power project resulted into the cancellation of the tender which delayed its completion upto 2012. Allegations of misuse of public funds by the Chinese company contracted to construct Karuma Dam on Karuma Falls on River Nile has greatly hindered the completion of this project.
- The insecurity/instability which has characterised Uganda for sometime prevented long term investment necessary for electric power development. In places like Gulu, Pader, Kitgum, the activities of the Lord's Resistance Army rebels prevented the extension of Hydro electric power in these areas.
- The dispersed nature of settlement in many rural areas which makes power transmission difficult in areas like Kotido, Luwero and Nakasongola has affected the government policy of supplying power to those areas.
- Remoteness of some areas due to the nature of relief , forests , swamps and lakes hinders the transmission of hydro electricity. For example Kalangala district and Buvuma district are up to date not connected to the national power grid because of inaccessibility caused by Lake Victoria.
- Conflict with environmentalists has delayed the exploitation of energy resources in Uganda. In 2019 the project to construct a power dam at Murchison falls on River Nile in

Masindi was suspended due to harsh criticism and demonstrations by environmentalist because of the effects it would bring on the River Nile hydrological ecosystem.

Geothermal energy in Bundibugyo is not yet exploited because it's located in Sempaya national park. In addition the exploitation of petroleum in Albert Flats met resistance from conservationists that it would affect wildlife.

- Illegal connections and power theft affecting power distribution. This has resulted into loss of money to companies like UMEME. Illegal connections are rampant in areas of Kisenyi, Katwe, Bwaise all in Kampala.
- There is limited domestic market for some energy resources due the high poverty levels. For example , very few people in rural areas eg in Kotido, Kaabong, Yumbe etc can't afford to use Hydro electric power since it requires periodical payments.
- The fluctuating prices of some energy resources account for the low levels development of the energy sector. For example all the petroleum products used in Uganda are imported from countries like Saudi Arabia, Libya and china. Therefore increased prices for these products lower consumption of car fuel, Diesel generators at Namanve and in Kalangala.
- High energy loss through transmission. This has mainly affected the transmission of hydro electric power. This is due to the use of outdated technology and dispersed settlement. This strains the machines at Bujagali, Nalubaale and Kiira Dams in Jinja to produce large quantities of power to cater for the power lost during transmission.
- The colonial treaties like the Nile Use Treaty and Low Rates Treaty have delayed the establishment of hydroelectric power plants along the Nile. The Nile treaty for example gives powers to countries like Egypt and Sudan to monitor the flow of water from Lake Victoria and River Nile. This has affected and delayed dam construction of dams like Karuma Dam in Kiryandongo and Isimba Dam in Kayunga due to bureaucracies involved in acquisition of permission from the member countries in the Nile Basin organization.
- Poorly developed transport routes limits the distribution of energy resources to upcountry like petroleum products to areas like Kaabong, Bundibugyo and Kapchorwa.
- Profit repatriation since the energy sector in Uganda is dominated by foreign based firms, for example the marketing and distribution of hydro electricity from Bujagali Dam in Jinja is done by UMEME, which is a foreign based company. This has hindered rapid expansion of the energy sector since foreign firms are more interested in how much they send back to their home countries.
- Limited research due to limited government support and inadequate capital has delayed the extraction of petroleum in Hoima and Buliisa. It has also delayed the exploitation of geothermal energy at Sempaya Hotsprings in Bundibugyo.

MEASURES BEING TAKEN TO DEVELOP THE ENERGY SECTOR IN UGANDA.

1. The government is promoting the rural electrification program. More areas are being connected to the national power grid. For example hydroelectric power has been extended to places like Bundibugyo, Ssembabule and Kapchorwa under this program.
2. There is construction of new dams to generate hydroelectric power. These include Isimba Dam in Kamuli completed in 2019 and the ongoing construction of Karuma dam in Kiryandongo all on River Nile. Other recently concluded dams include Mubuku III power station in Kasese, Nyagak power station in Zombo etc.
3. Research is being carried out by the ministry of energy and other private companies. For example , exploration of Oil in the Albert Flats by Tullow oil and Gas limited. Several explorations on Geo thermal Energy is underway in Kibiro Hotsprings in Hoima, Ssempaya and at Buranga Hotsprings in Bundibugyo by Gids consult and Moto Geothermal Project.
4. Afforestation and re-afforestation programs are being promoted so as to supply firewood and charcoal on a sustainable basis. Such programs are being promoted in areas like Mubende, Masaka, Jinja Mbale etc.
5. There has been liberalisation and privatisation of the energy sector. Foreign investors have been attracted with capital and skills in energy development like Tullow Oil and gas limited dealing in oil exploration in the Albertine Graben, Tronder power firm from Norway that developed the Bugoye power station in Kasese. Also distribution of power from Bujjagali Dam in Jinja is done by UMEME a South African firm.
6. There is use of energy saving gadgets in kampala, Jinja, Masaka, Mbarara and Mbale. Industries and homes use energy saving stoves to maximize the use of charcoal and firewood. Energy saving bulbs with a low consumption of hydro electricity are being used in Kampala, Gulu, Hoima etc.
7. There is diversification of the energy sources to supplement hydroelectric power like use of solar energy in Kampala and Masaka. There is also use of biogas from agricultural wastes in areas like Mbarara, Kabarole, Kanungu etc.
8. Government and other agencies are fighting corruption through the Parliamentary Accounts Committee (PAC), IGGs office , state house anti corruption committee etc. For example PAC on several occasions has intervened to assess the performance of UMEME, a company charged with the responsibility of distributing hydroelectric power in different parts of the country like Kampala, Mbale etc. In 2013 the IGG intervened over the allegations that the tendering process for the establishment of Karina dam on River Nile in Kiryandongo was fraudulently done.
9. Government has ensured political stability / security in areas like Kasese, Gulu , Kaabong and Kotido. This has promoted increased investment in the energy sector. For example the prevailing security in Kasese has facilitated the construction of hydro power stations on River mubuku like Mubuku III power station in Kasese.

10. There is fighting of illegal connections of HEP by UMEME helped with Uganda police in areas of Kampala, Wakiso and Mukono. Those caught are prosecuted in courts of law and later imprisoned.
11. The public is being sensitized on the sustainable use of energy resources. UMEME, the firm dealing in the distribution and marketing of hydro electricity has sensitized people especially in Urban areas such as Kampala, Wakiso and Mukono where the rate of energy consumption is high. This is done through the use of media like Print media, radios and Televisions like BBS and NBS.
12. There is importation and use of modern technology to facilitate the generation and distribution of energy. For example derricks have been imported to facilitate the exploration and exploitation of Oil in Hoima and Buliisa. Solar panels have been imported into the country and have been installed in a number of places like health facilities and schools in Masaka, Mbarara Gulu, Arua etc.
13. There has been acquisition of capital/ loans from international institutions to facilitate the generation of hydroelectric power and establishment of solar firms like at Kabulasoke in Gomba. For example world Bank gave a loan to facilitate the generation of hydroelectric power at Bujjagali on River Nile in Jinja.
14. There is training of manpower or labour to facilitate the effective exploitation of energy resources in Uganda. For example workers are trained at Kigumba Institute Of Petroleum Studies in Masindi to promote sustainable exploitation of petroleum in the Albertine Graben.
15. There is improvement and rehabilitation of transport routes to facilitate the transportation of energy resources for example the construction of Bundibugyo Road has promoted the transportation of petroleum products to Bundibugyo.
16. UMEME has promoted the use of prepaid digital metres (YAKA) to reduce on misappropriation of electricity payments by different officials. This is being done in Kampala, Jinja, Masaka etc.
17. Uganda has maintained a positive relationship with countries that form the Nile Treaty, that is Egypt, Sudan and South Sudan. This relationship has enabled Uganda to sustainably construct new hydroelectric power dams on River Nile like Isimba Dam in Kayunga and Karuma Dam in Kiryandongo.

CONTRIBUTIONS OF THE ENERGY SECTOR TO THE DEVELOPMENT OF UGANDA.

Positives.

- The energy sector is a source of employment opportunities . As a result people earn income which they use to meet their basic necessities hence improving on their standards of living. People are employed as engineers in the construction , generation

and transmission of hydro electric power in Jinja. For example over 500 people are employed in the construction of Karuma Dam in Kiryandongo.

- Generation of Hydroelectric Power in Uganda has promoted industrialization (manufacturing sector) in Uganda. Power is used to run machines, preservation of raw materials and finished products and processing raw materials.
- The energy sector has facilitated the extraction and processing of mineral resources. For example the generation of Hydroelectric Power at Mubuku III power plant in Kasese has facilitated the processing of Cobalt in Kasese.
- It is a source of foreign exchange to the country through the exportation of power to neighbouring countries like Kenya and Tanzania. This foreign exchange is used to develop other sectors of the economy. For example power from Nalubaale Dam on River Nile in Jinja is exported to Kenya, Tanzania and Eastern DRC.
- The exportation of power to regional countries from Nalubaale power plant on River Nile to Kenya, Tanzania and Rwanda has promoted regional cooperation. This has promoted international trade between Uganda and her neighbours.
- The energy sector has promoted education and research. For example many students visit Mubuku hydro power plant in Kasese for field work. As a result , students acquire new information and skills about power generation.
- The generation of of Hydroelectric power has enabled the construction of roads like the Nalubaale and Kiira Dams have enabled the construction of of a road linking Jinja to Njeru in Buikwe.
- The energy sector is a source of revenue to government through taxation and licencing of companies. For example companies like Tullow Oil and Gas limited which carries out Oil exploration in Hoima , UMEME firm responsible for marketing and distribution of power to all towns in the country, and workers in this sector pay huge sums of money in form of taxes to the government of Uganda. The revenue got is used by the government to provide social services to people for example schools.
- It has diversified the economy of Uganda by acting as an alternative source of income to government besides the revenue got from Agriculture. Forexample the income got from exportation of power from Nalubaale Dam in Jinja to Kenya and Tanzania relieves the government the burden of depending on few sectors particularly agriculture.
- It has promoted urbanisation hence bringing services closer to people. Forexample the generation of Hydroelectric power at Kiira and Bujjagali Dams in Jinja facilitated the development of Jinja into a city. As power as extended to areas businesses begin to grow facilitating urban development.
- It is a source of market to other sectors like forestry and Industrial sectors. Forexample the construction of hdro electric plants at Bujjagali in Jinja and Karuma Dam in Kiryandongo on River Nile provided market to the cement manufacturing industries like

Hima in Kasese and Tororo cement in Tororo. Also Budongo Forest in Masindi and Mabira Forest in Buikwe have provided poles for transmission of Hydroelectric power to different areas.

- It has promoted the transport sector , forexample petroleum products like Petro and diesel used to power vehicles in Kampala and other towns.
- The energy sector is used domestically in cooking , lighting etc. Forexample 92% of the population uses firewood and charcoal for cooking. Solar and HEP are used to light houses in Kampala, and other towns.

Negatives.

1. There is profit repatriation since the energy sector is dominated by foreign companies. This has hindered the rate of economic growth in the country. For example , UMEME a South African company charged with the distribution and marketing of Hydroelectric power repatriates profits to South Africa leaving little for domestic investment.
2. The energy sector has led to loss of vegetation cover through deforestation to get woodfuel (Firewood and charcoal). 92% of Uganda's population depends on biomass leading to the destruction of a number of forests like Mabira in Buikwe.
3. The energy sector has led to loss of lives due to accidents in the course of generating and transmitting hydro electric power at Jinja , Kampala etc. In some cases people have died of electric shock due to poor connections, illegal connections and poor handling of electric appliances especially in slums of kampala like Kikoni , Bwaise etc.
4. It has resulted into pollution of the environment especially the use of firewood in Masaka, Nakasongola, etc has resulted into air pollution due to smoke emitted to the atmosphere. The exploitation of petroleum in Hoima, Kikuube and Buliisa has resulted into land pollution. The establishment of Bujjagali , Nalubaale and Karuma Dams have led to water pollution of River Nile due to oil spills from turbines hence affecting water quality and acqautic life.
5. The establishment of energy generating plants has resulted into the displacement of people and their property. Like the exploration of petroleum in Hoima led to displacement of people along the shires of Lake Albert. The establishment of a thermal generator in areas of mutundwe displaced some people because of the unbearable noise.
6. It has led to development of urban centres with their associated problems like high crime rates, prostitution , unemployment etc.
7. It has led to competition with other sectors of the economy for capital , labour and space.

8. There is increased loss of government revenue through importation of petroleum products. Petroleum products are expensive and purchased on foreign currencies like Us dollars and Euros all which drain Uganda's coffers.